Homework 7

Alejandro De La Torre
MATH 421 — The Theory of Single Variable Calculus
Section 003
Instructor: Grace Work

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Collaboration Statement

Problem 1 *(Ross Chapter 2 11.4)*

Repeat Exercise 11.2 for the sequences:
$w_n = (-2)^n$, $x_n = 5^{(-1)^n}$, $y_n = 1 + (-1)^n$, $z_n = n\cos(\frac{n\pi}{4})$.

Question 11.2 asks you to do the following for each sequence:

(a) For each sequence, give an example of a monotone subsequence.

(b) For each sequence, give its set of subsequential limits.

(c) For each sequence, give its $\limsup$ and $\liminf$.

(d) Which of the sequences converges? diverges to $+\infty$? diverges to $-\infty$?

(e) Which of the sequences is bounded?

Discussion.
So we will analyze each sequence one by one, finding monotone subsequences (which we know must exist by 11.4 Theorem), subsequential limits, lim sup and lim inf, convergence/divergence, and boundedness for each.

For review, here are the relevant definitions:

(a) **Monotone subsequence**: A subsequence that is either non-increasing or non-decreasing.

(b) **Subsequential limits**: The set of all limits of subsequences of a given sequence.

(c) **Limit superior (lim sup)**: The largest subsequential limit of a sequence.

(d) **Limit inferior (lim inf)**: The smallest subsequential limit of a sequence.

Answers.
For $w_n = (-2)^n$:

(a) Monotone subsequence: All the even terms of the sequence form an increasing, monotone subsequence n is even, w_{2k} such that we have the terms: $(4, 16, 64, \dots)$. Alternatively, all the odd terms of the sequence form a decreasing, monotone subsequence n is odd, w_{2k+1} for $k \in \mathbb{Z}^+$ such that we have: $(-2, -8, -32, \dots)$.

(b)

$\square$

Problem 2 *(Ross Chapter 2 11.8)*

Use Definition 10.6 and Exercise 5.4 to prove $\liminf s_n = -\limsup(-s_n)$ for every sequence (s_n).
Discussion.

10.6 Definition:
Let (s_n) be a sequence in $\mathbb{R}$. We define:

$$\limsup s_n = \lim_{N\to\infty} \sup\{s_n : n > N\}$$

and

$$\liminf s_n = \lim_{N\to\infty} \inf\{s_n : n > N\}$$

5.4 Exercise:
Let S be a nonempty subset $\mathbb{R}$, and let $-S = -s : s \in S$. Prove $\inf S = -\sup(-S)$. *Hint: For the case $-\infty < \inf S$, simply state that this was proved in Exercise 4.9.*

Proof.
From the definition of 10.6, we have:

$$\liminf s_n = \lim_{N\to\infty} \inf\{s_n : n > N\}$$

where we describe the set $S_N = \{s_n : n > N\}$ such that we can rewrite the expression for $\liminf(s_n)$ as:

Thus, we have:

$$\liminf s_n = -\limsup(-s_n)$$

which applies for every sequence (s_n) since we know the result from Exercise 5.4 holds for any nonempty subset of $\mathbb{R}$, and thus every sequence. $\square$

Problem 3 *(Ross Chapter 2 11.10)*

Let (s_n) be the sequence of numbers in Fig. 11.2 listed in the indicated order.

(a) Find the set S of subsequential limits of (s_n).

(b) Determine $\limsup s_n$ and $\liminf s_n$.

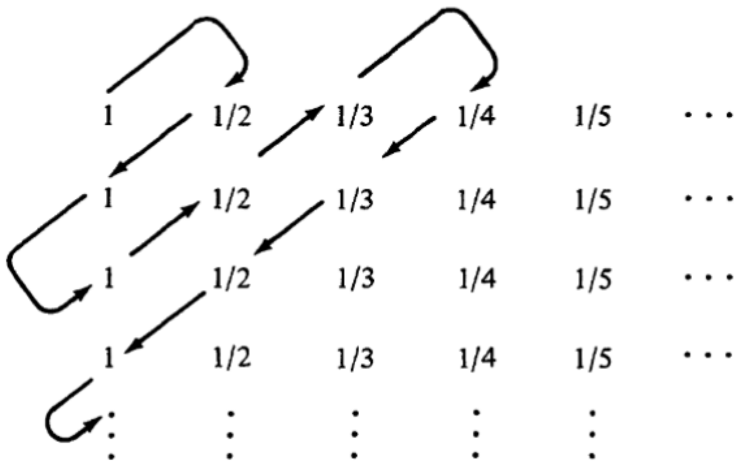

FIGURE 11.2

Discussion.

Proof.

□

Problem 4 *(Ross Chapter 2 12.2)*

Prove $\limsup |s_n| = 0$ if and only if $\lim s_n = 0$. **Discussion.**

Proof.

□

Problem 5 *(Ross Chapter 2 12.3)*

Let (s_n) and (t_n) be the following sequences that repeat in cycles of four:

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, ...)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, ...)$$

Find

(a) $\liminf s_n + \liminf t_n$

(b) $\liminf(s_n + t_n)$

(c) $\liminf s_n + \limsup t_n$

(d) $\limsup(s_n + t_n)$

(e) $\limsup s_n + \limsup t_n$

(f) $\liminf(s_n t_n)$

(g) $\limsup(s_n t_n)$

Discussion.

Proof.

$\square$

Problem 6 *(Ross Chapter 2 12.10)*

Prove (s_n) is bounded if and only if $\limsup |s_n| < +\infty$.

Discussion.

Proof.

$\square$